Monday lab 28/10/24

1. QUESTION: Maximum Sum Submatrix
   Find the maximum sum of a submatrix of size k × k in a given matrix.
   Example Matrix:
   1 2 3 4
   5 6 7 8
   9 10 11 12
   13 14 15 16
   For k = 2, the answer should be 52 (sum of submatrix [11,12; 15,16])

2. QUESTION: Row with Maximum 1s
   Given a boolean matrix where each row is sorted, find the row with maximum number of 1s.
   Example Matrix:
   0 0 1 1
   0 1 1 1
   0 0 0 1
   0 0 0 0
   Answer: Row 1 has maximum 1s (3 ones)

3. QUESTION: Print Matrix in Snake Pattern
   Print matrix elements in snake pattern.
   Example Matrix:
   1 2 3
   4 5 6
   7 8 9
   Output: 1 2 3 6 5 4 7 8 9
4. Given an n x n 2D matrix representing an image, rotate the image 90 degrees clockwise.
   Example
   Matrix = [[1, 2, 3],
   [4, 5, 6],
   [7, 8, 9]]
   Output: [[7, 4, 1],
   [8, 5, 2],
   [9, 6, 3]]
5. Given two matrices, A and B, matrix multiplication is performed to produce a new matrix C.
   Example: Input: Matrix A: 1 2 3
   4 5 6
   Matrix B: 7 8
   9 10
   11 12
   Output : Matrix C: 58 64
   139 154